REY AUDIE S. ESCOSIO

@ rsescosio@gmail.com @ rsescosio@alunos.fc.ul.pt Lisbon, Portugal
External links: Linkedin · Google Scholar · ResearchGate · GitHub · ORCID
Scopus · Web of Science ResearcherID · Ciência Vitae

BIOSKETCH

Rey (**he/him**) finished his B.Sc. Applied Physics and M.Sc. Applied Mathematics from the Institute of Mathematics, University of the Philippines Diliman. Since 2013, he has received two DOST scholarships – the DOST-SEI RA 7687 program, and the DOST-ASTHRDP. A portion of his graduate thesis on gradient-based optimization with acceleration and local search and its applications was awarded a student poster prize at the Society of Mathematical Biology 2020 Annual Meeting.

He is a Ph.D. student and research fellow of the FCT-funded project titled *"Targeted Control of COVID-19 Post Mass Vaccination"* of the Faculdade de Ciências, Universidade de Lisboa (FCUL). He is also affiliated with the BioSystems and Integrative Sciences Institute, FCUL and the Infectious Disease Modeling Group, University Medical Center Utrecht, Utrecht University.

Research interests:
infectious disease modelling | mathematical biology | optimization | complex systems | data science

EDUCATION

Doutoramento em Física e Astrofísica (Ph.D. Physics and Astrophysics) (Physics)

Faculdade de Ciências, Universidade de Lisboa

Jan 2024 – Present | Lisbon, Portugal

Project: Targeted Control of COVID-19 Post Mass Vaccination
- Focused on the research fellowship under the supervision of Dr. Ganna Rozhnova and Dr. Ana Nunes
- Researcher under the BioSystems and Integrative Sciences Institute, FCUL
- Junior researcher in the Infectious Disease Modeling Group, University Medical Center Utrecht, Utrecht University

Master of Science (Applied Mathematics) (Mathematics in Life and Physical Sciences)

Institute of Mathematics, University of the Philippines Diliman

Aug 2018 – Jan 2021 | Quezon City, Philippines

- Department of Science and Technology-Accelerated Science and Technology Human Resource Development Program - National Science Consortium (DOST-ASTHRDP) Scholarship awardee

Thesis: *A perturbed accelerated gradient descent algorithm using an n-dimensional golden section search method*
- Focused on gradient-based optimization with a local search applied to benchmark functions, matrix decomposition, parameter estimation of mathematical models, signal processing, and single-layer perceptron
- Student researcher under the supervision of Dr. Renier Mendoza (Jan 2019 – present) for my graduate thesis and presented my research at various international and national conferences. Initial preprint accessible here.

Bachelor of Science in Applied Physics (Instrumentation Physics)

National Institute of Physics, University of the Philippines Diliman

Jun 2013 – Jun 2018 | Quezon City, Philippines

- Department of Science and Technology-Science Education Institute (DOST-SEI) RA 7687 Scholarship awardee

Thesis: *Response of single and connected Hodgkin-Huxley neurons to a sinusoidal input*
- Worked on varying stimuli to coupled differential equations describing neural response analyzed dynamically and statistically in different cases
- Student researcher in Complexity Science Group under the supervision of Dr. Johnrob Bantang (Oct 2015 – May 2018)
- Presented in three Samahang Pisika ng Pilipinas (SPP) Physics Conferences (2016 – 2018) accessible on my Google Scholar profile.

RESEARCH

Research publications

1. Canfora, B., **Escosio, R. A. S.**, et al., (2025). Counterfactual evaluation of elementary and secondary school policies in the COVID-19 pandemic. *medRxiv*. (Shared first co-authorship; under review).
2. Canfora, B., **Escosio, R. A. S.**, et al., (2025). Retrospective evaluation of school-related measures on pre-vaccination transmission dynamics of SARS-CoV-2. *medRxiv*. (Shared first co-authorship; under review).
3. Mata, M. A. E., **Escosio, R. A. S.**, et al., (2024). Analyzing the Dynamics of COVID-19 Transmission in Select Regions of the Philippines: A Modeling Approach to Assess the Impact of Various Tiers of Community Quarantines. *Heliyon*. (Shared first co-authorship).

4. **Escosio, R. A. S.**, et al., (2023). A Model-Based Strategy for COVID-19 Vaccine Roll-out in the Philippines. *Journal of Theoretical Biology*.
5. **Escosio, R. A. S.**, & Mendoza, R. G., (2022). A Modified Accelerated Gradient Descent Using an N-Dimensional Golden Section Search for Escaping Saddle Points. *SSRN*. (Preprint).

International conference presentations

1. **Escosio, R. A. S.**, Canfora, B., Boldea, O., Nunes, A., Rozhnova G., *Quantifying the role of age in COVID-19 transmission.* 8th International Workshop on Mathematical Biology (IWOMB 2026), in-person meeting, January 6-9, 2026. (poster presentation)
   - Won the student poster prize, 2nd place
2. **Escosio, R. A. S.**, Canfora, B., Boldea, O., Nunes, A., Rozhnova G., *Quantifying the role of age in COVID-19 transmission.* 10th International Conference on Infectious Disease Dynamics (EPIDEMICS 10), in-person meeting, November 30-December 3, 2025. (poster presentation)
3. **Escosio, R. A. S.**, Canfora, B., Boldea, O., Nunes, A., Rozhnova G., *Characterizing Sociodemographic Influences to the Reproduction Number of SARS-CoV-2.* 16th Dynamical Systems applied on Biology and Natural Sciences (DSABNS 2025), in-person meeting, January 20-24, 2025. (poster presentation)
4. **Escosio, R. A. S.**, Cawiding, O. R., Vergara, T. H. M., et al., *The What-Ifs of Early COVID-19 Transmission in the Constituent Units of the National Capital Region, Philippines.* Society of Mathematical Biology 2022 Annual Meeting (SMB 2022), hybrid meeting, September 19-23, 2022. (online poster presentation)
5. **Escosio, R. A. S.**, and Mendoza, R. G., *A Modified Accelerated Gradient Descent Using an n-Dimensional Golden Section Search for Escaping Saddle Points.* The 8th International Conference on Control and Optimization with Industrial Applications, virtual attendance, August 24-26, 2022. (oral presentation)
6. **Escosio, R. A. S.**, de los Reyes, A. A. V, Mendoza, V. M. P, & Pilar-Arceo, C. P. C. *Modelling COVID-19 Dynamics with Different Community Quarantine Protocols in National Capital Region, Philippines.* Society of Mathematical Biology 2021 Annual Meeting (eSMB 2021), virtual meeting, June 13-17, 2021. (poster presentation)
7. **Escosio, R. A. S.**, & Mendoza, R. G., *Parameter Estimation of the Fitzhugh-Nagumo Model via a Perturbed Accelerated Gradient Descent Algorithm with an n-Dimensional Golden Section Search Method.* Society of Mathematical Biology 2020 Annual Meeting (eSMB 2020), virtual meeting, August 17-20, 2020. (poster presentation)
   - Won the subgroup student poster prize
   - Initially accepted for the 12th European Conference on Mathematical and Theoretical Biology (ECMTB 2020) but redirected to eSMB 2020

Participation

1. 6th International Workshop on Mathematical Biology (IWOMB 2023), Paradise Garden Resort Hotel and Convention Center, Boracay, January 22-25, 2024.
2. 5th International Workshop on Mathematical Biology (IWOMB 2023), University of the Philippines Baguio, July 26-27, 2023.
3. Unifying the Epidemiological and Evolutionary Dynamics of Pathogens Program, Nordic Institute for Theoretical Physics, May 29-June 23, 2023 (attended first two weeks).
4. Spring College on the Physics of Complex Systems, International Center for Theoretical Physics, Trieste, Italy, February 20-March 17, 2023
5. SEAMS School 2022 on Modern Trends in Signal and Data Processing, University of the Philippines Diliman, December 5-13, 2022.
6. 3rd International Workshop on Mathematical Biology (IWOMB 2020), University of the Philippines Los Baños, January 6-8, 2020.
7. ICTP Asian Network School and Workshop on Condensed Matter and Complex Systems 2019, University of the Philippines Diliman, November 4-8, 2019.
8. 16th International Conference on Molecular Systems Biology (ICMSB 2019), De La Salle University, October 28-31, 2019.
9. 2nd International Workshop on Mathematical Biology (IWOMB 2019), Bohol Bee Farm, January 6-10, 2019.

EXPERIENCE

University Research Associate, COVID-19 Modelling Project

Natural Sciences Research Institute, University of the Philippines Diliman

Jan 2022 – Dec 2022 Quezon City, Philippines

- Responsible for the model formulation, preprocessing of data, programming, science writing, and other admin duties
- Worked under Dr. Carlene P.C. Pilar-Arceo and the Modelling and Applications Group of the Institute of Mathematics

---

Science Research Specialist, COVID-19 Modelling Project

Center for Applied Modeling, Data Analytics, and Bioinformatics for Decision Support Systems in Health

Nov 2021 – Dec 2021 | Davao City, Philippines

- Focused on parameter estimation for the COVID-19 modeling dynamics of the constituent units of the Davao Region
- Responsible for preprocessing data for parameters, programming, and automation of codes, and preparing presentations for the project
- Worked under Dr. May Anne E. Mata

Lecturer, Precalculus Courses

Institute of Mathematics, University of the Philippines Diliman

Sep 2021 – Jan 2022 | Quezon City, Philippines

- Taught precalculus to two online classes of early-stage university students

---

Research Associate, COVID-19 Modelling Project

Modelling and Applications Group, Institute of Mathematics, University of the Philippines Diliman

May 2021 – Nov 2021 | Quezon City, Philippines

- Focused on two projects regarding COVID-19 dynamics with objectives on infectious-specific and age-structured vaccination models.
- Responsible for handling datasets, researching methods, and preparing codes, presentations, and manuscripts for the projects
- Worked under coinvestigators Dr. Aurelio A. De los Reyes V, Dr. Carlene P.C. Pilar-Arceo, and Dr. Victoria May P. Mendoza

---

Project Assistant, COVID-19 Modelling Project

Resilience Institute, University of the Philippines

Nov 2020 – May 2021 | Quezon City, Philippines

- Mainly works with the Modelling and Applications Group of the Institute of Mathematics, UP Diliman, and other teams from UP Baguio, UP Mindanao, and UP Los Baños
- Assists in mathematical model building for specific interventions applicable to the National Capital Region and works further on model fitting, simulations, and projections for scenario analysis
- Worked under co-investigators Dr. Aurelio A. De los Reyes V, Dr. Carlene P.C. Pilar-Arceo, and Dr. Victoria May P. Mendoza

SKILLS

Programming languages and version control
Proficient in Python, MATLAB, Wolfram Mathematica, and R; familiar with Stan, C++, and SQL; familiar with the GitHub environment

Typesetting and other software
Effective in using LaTeX for documents, presentations, and so on (including this CV); proficient in MS Office and similar softwares and in Adobe Photoshop and other graphics editing software

Languages
English and Filipino – proficient in both oral and written communication

Interpersonal skills
Held leadership positions from my organizations and affiliations; thinks critically and creatively; communicates effectively; works efficiently with teams; fun to be with